

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

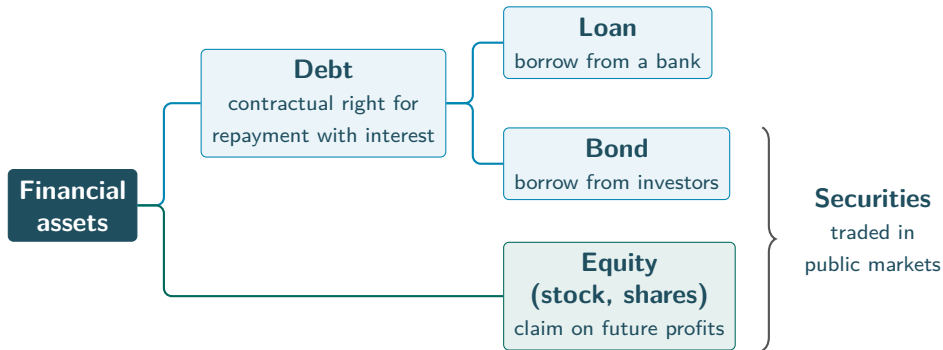
- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Every Financial Asset Is Debt or Equity

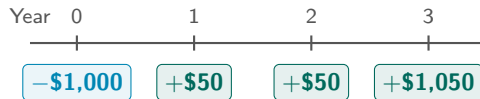
- **Financial asset**, value from a contractual right or ownership claim



A Bond's Cash Flow Is Fixed, a Stock's Is Not

Bond

\$1,000, 3 years, 5% interest

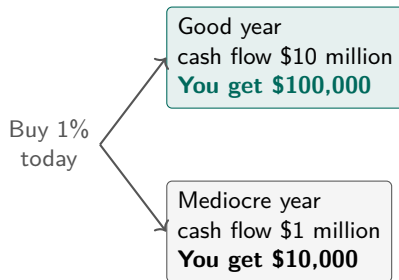


Every payment known in advance

Stock

1% of the company

Next year



Value of claim depends on how the company does

Common Types of Corporate Financing

- **Debt**, lending to the company
 - Main forms, bonds and loans
 - Cash flow, a fixed promised amount
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- **Equity**, stock ownership in the company
 - Cash flow, dividends (and buybacks)
 - “Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout
 - Voting rights

Common Types of Corporate Financing

	Debt	Equity
Payment	Fixed promised payment	Residual (whatever is left over)
Priority in bankruptcy	Highest	Lowest
Risk	Low	High
Expected return	Low	High
Voting and control rights	No	Yes
Term	Dated	Perpetual
Share of total U.S. corporate financing (market values)	About 25%	About 75%

Module 5, Part 1

Bonds

Bonds, Definitions

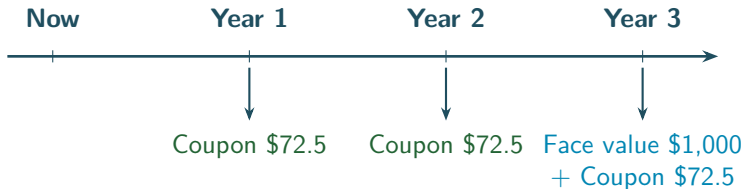
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- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$72.5 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
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- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$

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 - E.g., **Module 4**’s Puerto Rico zero-coupon bond, the coupon was 0 but the interest rate was 5.15%
 - Coupons can also be higher than the interest
- **Zero-coupon bonds**, bonds that pay the face value at maturity with no coupon payments along the way

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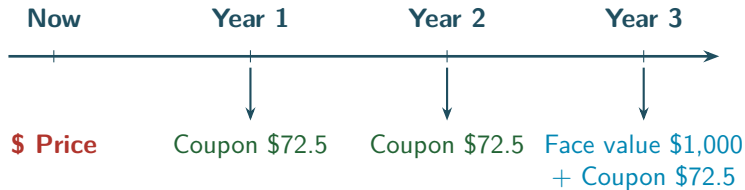
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- The required rate of return r is set by the opportunity cost, “what could I get investing in something else with a similar risk”
 - Depends on the time value of money and the riskiness of the bond (more on this a bit later)

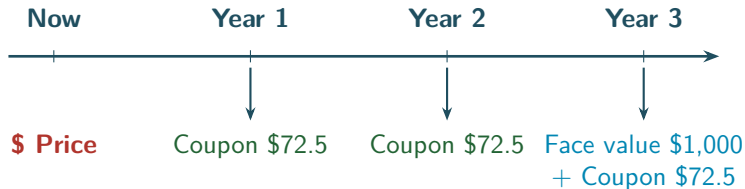
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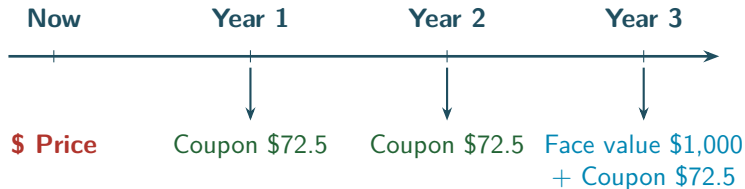
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$$\text{Price} = \frac{\$72.50}{(1 + 0.04)^1} + \frac{\$72.50}{(1 + 0.04)^2} + \frac{\$72.50 + \$1,000}{(1 + 0.04)^3} = \$1,090.19$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$72.5, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to price the bond

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- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

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- Solution,
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- The output is $-\$1,090.19$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the price of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate (the required rate of return r)

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The price (value) of a bond is the discounted payments, at the required rate of return

Bond Yields

- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price (also simply the "yield")

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- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price (also simply the "yield")
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield to maturity by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 7.25%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,090.19. What is its YTM?

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- Solution,
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 - -1090.19 **PV**
 - **CPT**, then **I/Y**
- The yield is 4%

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield to maturity
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield to maturity of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	−310	0	1,000
b. Bond B	?	9	−310	0	1,000
c. Bond C	11	8	?	0	−1,000

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

Practice Problem

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	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	428.88	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Practice Problem

- A firm's bonds pay an annual coupon rate of 6.8%. Their face value is \$1,000. Their price is \$795.69. Their yield to maturity is 9.50%
- What is the remaining maturity of these bonds?

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$

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- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**

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- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**
- The remaining maturity is about 14 years ($N = 13.98$)

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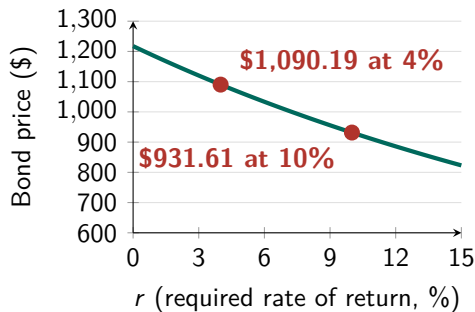
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 - Higher prevailing (risk-free) interest rate, higher r
 2. Risk of default (the opportunity cost for risk-taking)
 - Higher (systematic) risk of default, higher r

Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

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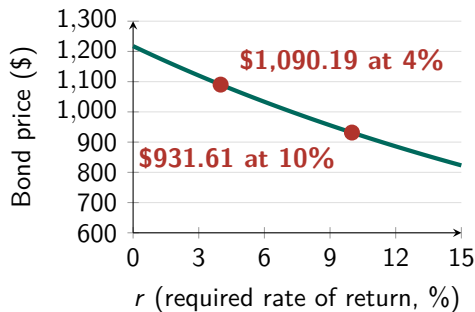


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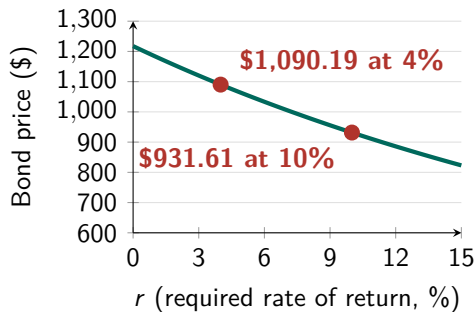


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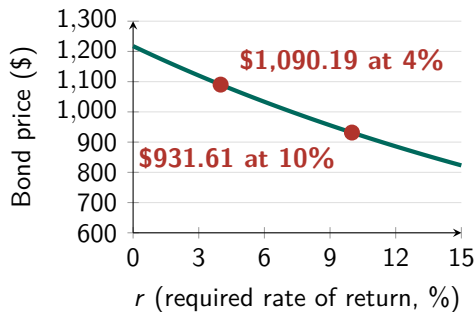


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- **Default premium**, the additional yield on a bond that investors require for bearing credit risk

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- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

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- Yields tend to be higher (prices lower) for lower-rated bonds

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- Interest rates and bond prices move in opposite directions
- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 2

Stocks

Goals

- **Part 1, bonds**
 - Differences between debt and equity
 - Valuation
 - Yields
 - Interest rates
 - Default risk
- **Part 2, stocks**
 - Valuation, dividend discount model
 - Valuation, comparables
 - Market efficiency

Valuing Stocks

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 1. Dividend discount model
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 1. Dividend discount model
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- Professionals use a combination
- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\frac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?

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- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

Valuing Stocks, Dividend Discount Model

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- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$ are *expected* dividends

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- If dividends are expected to grow at a constant rate g ,

$$P_0 = \frac{\text{Div}_1}{r_e - g}$$

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$$P_0 = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P_0 = \frac{\text{Div}_1}{r_e - g}$$

- The latter formula is often called the “**Gordon growth model**”

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
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- Stock value today, the present value of these cash flows,

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- That is, dividend yield plus expected capital appreciation (in %)

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- Uncertain dividend forecasts
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- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies
 - But future dividends are then probably even more uncertain!

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \dots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

Practice Problem

- Consider the following three stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
 - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$\begin{aligned} P_C &= \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} + \\ &\quad \frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6} \\ P_C &= \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ &\quad + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6 \\ P_C &= \$104.51 \end{aligned}$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity, $P_A = \$142.86$, $P_B = \$166.67$, and $P_C = \$156.50$, so Stock B is the most valuable

Valuing Stocks, Comparables

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

Example, Valuation by Comparables

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- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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- Assumes the comparables are priced correctly
 - Cannot tell if an entire industry is overvalued or undervalued
 - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

Buying and Selling Stock on the Stock Market

- Current stock price

FedEx Corporation (FDX)

☆ Add to watchlist

NYSE - Nasdaq Real Time Price. Currency in USD

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary

Chart

Conversations

Statistics

Profile

Final

Previous Close	249.02	Market Cap	64.318B
Open	246.00	Beta	1.57
Bid	239.98 x 100	PE Ratio (TTM)	22.11
Ask	240.10 x 100	EPS (TTM)	10.86
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)
Volume	2,359,358	Ex-Dividend Date	N/A
Avg. Volume	1,866,076	1y Target Est	284.69

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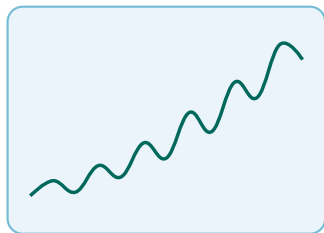
- Market capitalization
- **Beta**, a measure of risk (how much the stock moves with the overall market)
- Price/earnings
- EPS
- Dividend and dividend yield

Can You Beat the Stock Market?

- It's really hard!

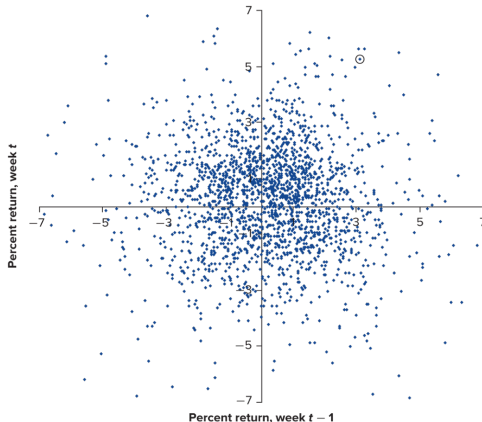
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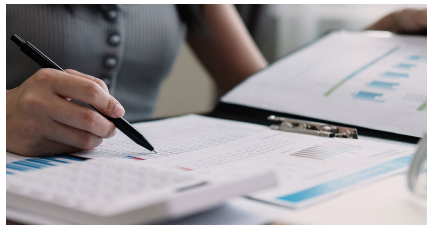
Discuss, Where Is the Pattern?

- Scatter plot of the NYSE Composite Index return in one week against its return in the previous week



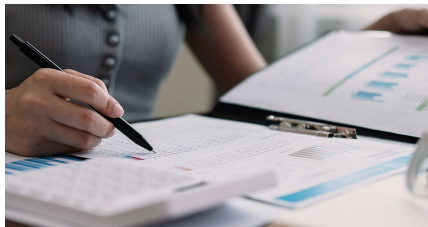
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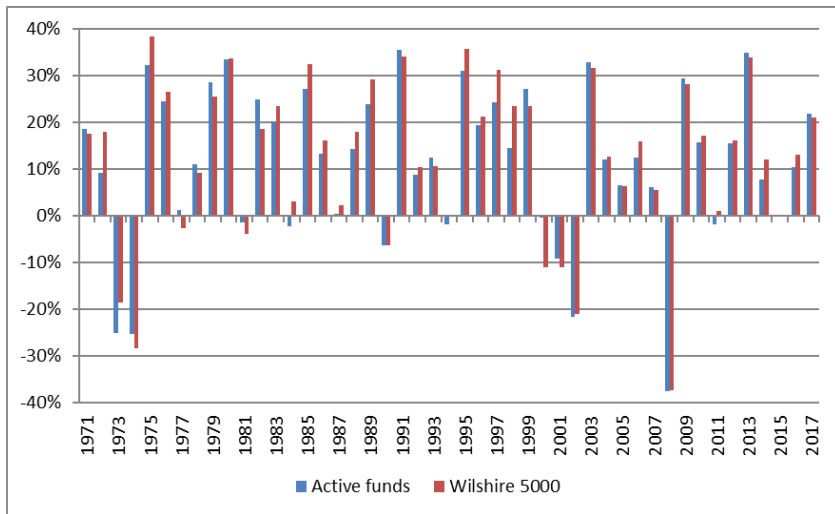


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Professional Fund Managers Don't Generally Beat the Index!



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- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”
- What **type** of information is captured in the stock price?
 1. All historical price information? (“weak” efficiency)
 2. All public, easily available information? (“semi-strong” efficiency)
 3. Also all private or difficult-to-interpret information? (“strong” efficiency)

What Does This Mean for Investors?

Pros 👍

- If stocks are fairly priced, even an investor who **knows nothing** can buy **any** stock and expect it to fairly compensate them for the risk they take

Cons 👎

- To *consistently* make money above the *fair* (*risk-compensating*) return by betting on publicly traded shares, an investor needs a very distinct competitive advantage
- E.g., expertise or access to information that only a few people can figure out

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- Constant dividends, $P_0 = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P_0 = \text{Div}_1 / (r_e - g)$
- One period, expected return = dividend yield + expected capital appreciation
- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Beating the market is really hard, prices already reflect available information (the efficient markets hypothesis)

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- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields
- Stocks, the dividend discount model or comparables, and efficient markets make the index hard to beat

Thank you!

Please let me know if you have any questions.